we are looking for $\frac{\partial loss}{\partial l_{ij}}$

$$loss = \left(\sum_{i=1}^{batch\_size} logprobs_{i y_i} \right) \frac{1}{batch\_size}$$

where $Y = [y_1, \dots, y_{batch\_size}]$: index of the correct choice according to the sample in the row i

$$\Rightarrow loss = -\sum_{i=1}^{batch\_size} \ln(probs_{i y_i}) \cdot \frac{1}{batch\_size}$$

$$= -\sum_{i=1}^{batch\_size} \ln\left(\frac{count_{i y_i}}{\sum_j count_{ij}} \right) \cdot \frac{1}{batch\_size}$$

$$= \frac{-1}{batch\_size} \sum_{i=1}^{batch\_size} \ln\left(\frac{e^{l_{i y_i} - \max_k(l_{ik})}}{\sum_j e^{l_{ij} - \max_k(l_{ik})}} \right)$$

~~we can simplify the max~~

$$= \frac{-1}{batch\_size} \sum_{i=1}^{batch\_size} \ln(e^{l_{i y_i}}) - \ln\left(\sum_j e^{l_{ij}}\right)$$

~~=> if j ∈ Y~~

$$\Rightarrow \frac{\partial loss}{\partial l_{ij}} = \begin{cases} \left(-1 + \frac{e^{l_{ij}}}{\sum_k e^{l_{ij}}}\right) \frac{1}{batch\_size} & \text{if } j = y_i \\ \frac{-e^{l_{ij}}}{\sum_k e^{l_{ij}}} \cdot \frac{1}{batch\_size} & \text{else.} \end{cases}$$